



Noroff

School of technology
and digital media

Technical Report

Community Science Museum

Mona Dyrhol

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Table of Contents

1. Summary	3
2. Body	5
2.1. Introduction	5
2.2. Main section of report	5
2.3. Conclusion	9
3. References	11
4. Acknowledgements.....	13
5. Appendices.....	14



1. Summary

This project contained of designing and developing a fully responsive and accessible website for the Community Science Museum.

The goal was to create a website that encouraged children to visit the museum in person with their families through an informative, engaging, accessible, and visually appealing website. The website is accessible from a working navigation menu that works on both desktop and mobile.

The focus was on good SEO ranking, responsiveness, accessibility and user experience across different devices and browsers. The website was developed using semantic HTML and CSS, Flexbox and CSS Grid.

Throughout development, different accessibility and performance testing tools were used, such as WAVE Accessibility, W3V Validator, Google Lighthouse and keyboard navigation testing. These tools helped to identify and resolve issues related to heading hierarchy, focus states, image accessibility, HTML validation and performance. To ensure responsiveness different units were used, such as aspect-ratio, clamp() and vw.

The outcome of this project consists of multiple pages, interactive navigation elements, responsive layout and structured content sections. Design choices for the project were focused on readability, consistency, accessibility, and visual hierarchy while maintaining a modern, engaging and appealing user interface.

This project shows a good understanding in WCAG accessibility principles, responsive design, and testing



methodologies. The project also shows balance between aesthetics and usability.



2. Body

2.1. Introduction

The purpose of this project was to design and build a fully responsive website for a science museum that works from a navigation menu and is engaging and visually appealing.

The website provides information about the museum such as, exhibitions, events, explore, how to get in touch, and how to get involved for a young audience and it's family.

The website is easy to navigate through a navigation menu, and is meant to inspire visitors to visit and explore the museum in person.

2.2. Main section of report

Layout and Design:

The target group for this project is young children aged 7 to 15 with families. The design choices was made to engage and inspire visitors to explore the museum in person.

The design focuses on usability, clarity and accessibility. The layout was made clean and simple to ensure visitors can easily navigate and understand the content.

For the colours scheme the choices was made to create a calm, engaging and professional appearance that is suitable for a science museum and target the audience. To improve and ensure readability there is high contrast between text and background.



Images are placed using grid layout and styled with properties such as aspect-ratio, object-fit and width to maintain consistency and responsiveness. They are placed to support the content and create visual interest.

The typography was chosen to be clear and readable across different devices. The font sizes is adjusted using responsive units to ensure consistency on different devices.

A important part of the design was to ensure it was responsiveness. To ensure good user experience the website was developed from a mobile-first view. Media queries were used to adjust the layout from smaller screens to larger.

On smaller screens content is aligned vertically, while on larger screens content is arranged side by side.

HTML semantic elements, descriptive links, aria labels and alt text for images was used to ensure accessibility. To ensure accessibility from the navigation menu with checkbox, tabindex and outline was used to be keyboard navigable.

CSS Grid was used for the main layout because it allows precise control over placement of elements such as text and images. For navigation and smaller components Flexbox was used for better spacing and alignment.

Implementation:

The website is built with HTML5, CSS, Flexbox and CSS Grid.



The website is developed using semantic HTML elements such as <head>, <header>, <main>, <section>, <article>, <address>, <form>, and <footer>. Accessibility links are descriptive, e.g., mailto and tel links.

The semantic HTML improves the website accessibility and makes the content easier to understand for users and search engines.

In the head of each page is a unique meta title and description for better SEO.

All HTML pages is located in the root directory, while stylesheets and assets are organized into separate folders for better structure and maintainability.

Testing and evaluation:

Testing tools used for this project:

- Google Lighthouse (performance and accessibility)
- WAVE Accessibility Tool (accessibility testing)
- W3C Validator (HTML errors)
- Keyboard navigation testing

W3C Validator detected duplicated ID and issues with heading hierarchy. These were corrected to ensure valid HTML structure.

WAVE Accessibility tool highlighted missing alt attributes and duplicated form labels. After fixing error, the project achieved an AIM Score of 10/10 across all pages, with improved semantic HTML, ARIA support, and descriptive alt text.

Google Lighthouse were used to for performance, Accessibility, Best Practices and SEO testing. Desktop performance have a high score, while the performance



testing for mobile was lower due to the use of visual effects such as backdrop blur, layered overlays and images. To preserve the visual identity and user experience for the project, design choices were intentionally kept.

Accessibility, Best Practices and SEO have 100% score across pages. SEO on Homepage have a 91 score due to buttons called “see more”.

Results are included in the appendices.

The website was evaluated through full keyboard navigation testing. The testing identified missing focus states on multiple interactive elements. These issues were resolved by implementing focus indicators and adding them to hover states to ensure improved accessibility and user experience across website.

Deployment and maintenance:

The website was deployed using GitHub Pages. The deployment was done by pushing the code to a GitHub repository and enabling GitHub Pages from the repository settings.

To maintain the website, updates go through the GitHub repository. The website has a structured folder system and modular CSS, which can make updates to content more efficiently.

The project mainly uses a simple Git workflow.

The main branch contains the final and stable version of the project. Under development, a single branch was used for creating navigation bar and footer. Most development work was completed directly in main, with clear and precise commit messages to describe changes.



Challenges:

One of the main challenges was controlling the layout behaviour in CSS Grid, especially when positioning elements and managing different layouts across screen sizes. Other challenges was ensuring that images scaled and behaved correctly without breaking layout and making horizontal scroll.

Challenges were solved by using clear grid placement such as “grid-column”, “grid-row”, and responsive techniques as clamp(), flexible units (vw, %), and aspect-ratio.

2.3. Conclusion

This project have become a fully responsive, engaging and accessible website for the Community Science Museum for a young audience. The website includes 7 pages and a placeholder page for interactive features that required JavaScript. Across pages are interactive elements, and a structured layout designed to improve user experience and accessibility.

The design choices reflect a focus on accessibility, responsive design, and consistent user experience across different devices and input methods.

Focus throughout the development prosses was placed on semantic HTML, responsive design, accessibility, and testing using W3C Validator, WAVE accessibility tool, Lighthouse, and keyboard navigation. These test helped to identify and resolve issues related to contrast, heading structure, and keyboard navigation.



This project meets the initial requirements and shows a good understanding of frontend development principles. Some of the challenges that was encountered were in relation to cross-browser compatibility and imaging placement. For future improvements could include further performance enhancements and additional interactive feature.



3. References

Images used from unsplash in this project:

Hero homepage

<https://unsplash.com/photos/pink-and-white-balloons-on-white-chair-MldLSFHYuMk>

Exhibitions

<https://unsplash.com/photos/a-satellite-dish-sitting-on-top-of-a-hill-under-a-night-sky-filled-with-stars-N3IYNg4X73o>

<https://unsplash.com/photos/white-and-orange-robot-near-wall-yLcK3Itx6ok>

<https://unsplash.com/photos/school-of-fish-in-water-BJUoZu0mpt0>

<https://unsplash.com/photos/a-person-holding-a-frog-F3OSsb4IfmU>

<https://unsplash.com/photos/galaxy-wallpaper-OwbUG4UrWm0>

Events

https://unsplash.com/photos/birthday-decor-lot-MD_ha01Bk7c

Testing tools:



Google Lighthouse
WAVE
<https://validator.w3.org/>

Design:
<https://figma.com/>

Fonts:
<https://fonts.google.com/>

Resources:
<http://chatgpt.com/>



4. Acknowledgements

I would like to acknowledge the guidance and feedback that was provided by my course instructors throughout this semester. Their support was valuable in this project and helped improving the design and development of the website.

I would also like to acknowledge the course materials for their detailed explanations of the relevant topics. Online resources such as W3C Validator, Lighthouse, and WAVE, were helpful tools for debugging, accessibility improvements, and performance evaluation.

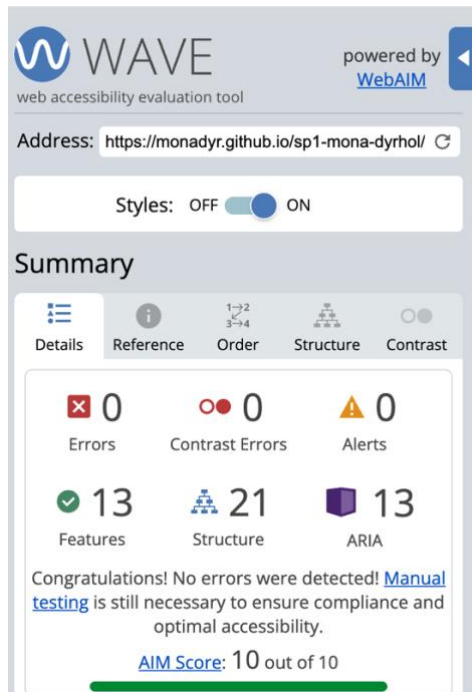
Additionally, the use of AI-assistance was used during the development process as a support resource.



5. Appendices

Appendix A

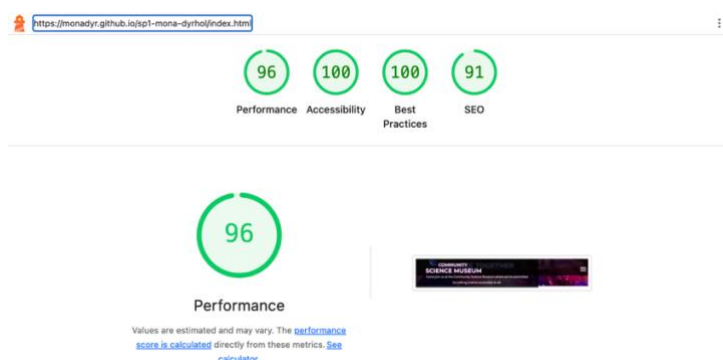
- WAVE result screenshot homepage

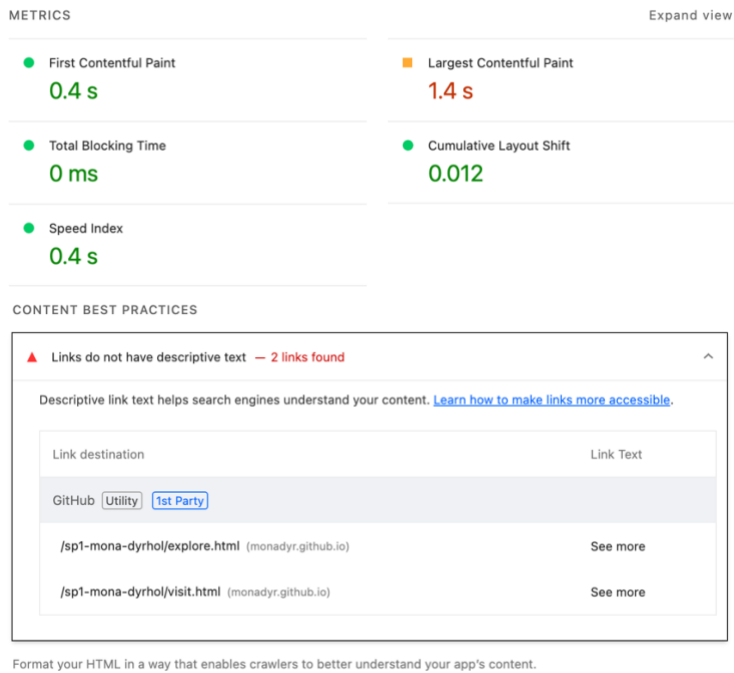


All pages was tested using WAVE Accessibility tool and achieved an AIM score of 10/10.

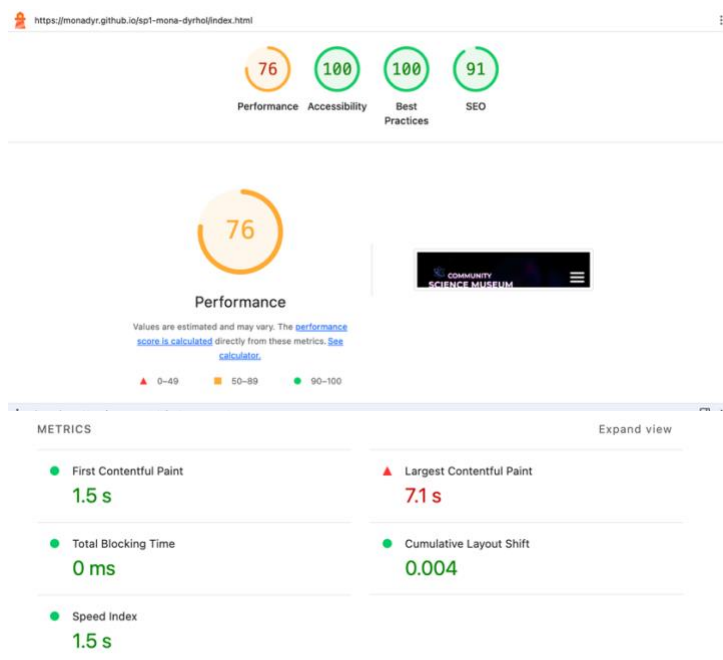
Appendix B

- Lighthouse homepage desktop result screenshot





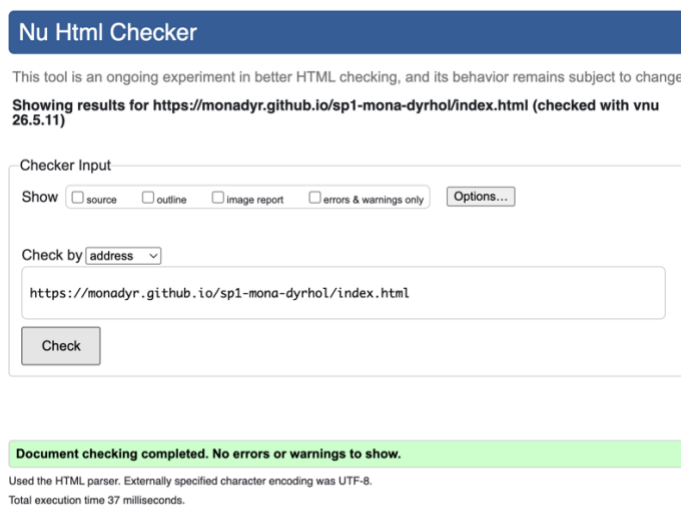
- Lighthouse homepage mobile result screenshot



Appendix C



- W3V Validator result screenshot



Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for <https://monadry.github.io/sp1-mona-dyrhol/index.html> (checked with vnu 26.5.11)

Checker Input

Show ☐ source ☐ outline ☐ image report ☐ errors & warnings only [Options...](#)

Check by [address](#) [source](#) [image](#)

<https://monadry.github.io/sp1-mona-dyrhol/index.html>

[Check](#)

Document checking completed. No errors or warnings to show.

Used the HTML parser. Externally specified character encoding was UTF-8.
Total execution time 37 milliseconds.

[About this checker](#) • [Report an issue](#)

All pages was tested using W3V Validator and have 0 errors or warnings.

Appendix D

- Fimga prototype, wireframe and styleguide link:

<https://www.figma.com/design/YkgqiXc5pQ9Pr5nEC8u7SH/Community-Science-Museum?node-id=13-195&t=9zVaK5FJMo1MGUW0-1>

